

Joshua Clavecillas

Joshclavecillas@gmail.com • (909) 782-0816 • California, USA
Physics • Applied Mathematics • California State University San Bernardino
GitHub • LinkedIn

Education

California State University, San Bernardino *2024 – Present*
B.S. Physics • B.S. Applied Mathematics (Double Major)

Chaffey College (Former) *2022 – 2024*
Physics Coursework (General Physics I–III, Differential Equations, Calculus I–III)

Research Interests

Condensed Matter Physics • 2D Materials (MoSe₂/BN)
Optical Neural Networks • Spectroscopy • Quantum/Computational Materials Science

Research Experience

Undergraduate Research Journal Project *2024 – Present*
CSUSB Department of Physics

- Developing a multi-section research journal focusing on ONN architectures and 2D TMD heterostructures.
- Implementing Python methods for peak-finding in photoluminescence spectra; drafted reusable code templates.
- Hands on table experience in Optical networks used for spectroscopy

Teaching Tutoring Experience

STEM Tutor (Math & Physics) *2022 – Present*

- Tutored students in Calculus, Linear Algebra, Classical Mechanics, E&M, and Thermodynamics.
- Designed problem sets, conceptual explanations, and exam-prep outlines for high-school and college students.
- Special focus on explaining entropy, Lagrangian mechanics, multivariable calculus, and spectrometry math.

Technical Skills

- **Programming:** Python (NumPy, Matplotlib), MATLAB, LaTeX, Mathematica
- **Lab/Instrumentation:** Spectrometers, photonics setups, oscilloscope, data acquisition
- **Math/Physics:** PDEs, Numerical Methods, Statistical Thermodynamics, Relativity, ODE solvers

Awards & Leadership

- STEM peer mentor for incoming physics majors